DLA - Development of the Nervous System - Part 1

From ectoderm to spinal cord and nerves

Objectives

SOM.MKII.BPM1.1.MSK.1.ANAT.1094	Describe the formation of the neural plate and define the term neuroectoderm
SOM.MKII.BPM1.1.MSK.1.ANAT.1095	Describe the major steps in formation of the neural tube and neural crest.
SOM.MKII.BPM1.1.MSK.1.ANAT.1096	List the derivatives of the neural tube.
SOM.MKII.BPM1.1.MSK.1.ANAT.1097	List the nervous system derivatives of the neural crest cells and, briefly list other tissues or organs that are formed from neural crest cells.
SOM.MKII.BPM1.1.MSK.1.ANAT.1098	Describe the three zones which form in the neural tube and explain how each of these zones contributes to formation of the spinal cord.
SOM.MKII.BPM1.1.MSK.1.ANAT.1099	Describe the anatomical sites of formation of somatic and visceral motor, and sensory neurons from neuroblasts.
SOM.MKII.BPM1.1.MSK.1.ANAT.1100	Describe the development of the meninges.
SOM.MKII.BPM1.1.MSK.1.ANAT.1101	Describe the embryological origin of the vertebrae and intervertebral discs.
SOM.MKII.BPM1.1.MSK.1.ANAT.1102	Describe the location of the primary and secondary ossification centers of a typical vertebra and give the time at which they occur.

Special note

Some slides have a green background, this will not be covered in the video recording but are put so you have the information available

The Big picture

Sperm → Zygote
Secondary oocyte → Zygote

Zygote → Morula → Blastocyst

Blastocyst → Outer Cell Mass → Trophoblast

Trophoblast → Syncytiotrophoblast
Trophoblast → Cytotrophoblast

Blastocyst → Inner Cell Mass → Embryoblast

Embryoblast → Hypoblast → Umbilical vesicle

Embryoblast → Epiblast

Epiblast → Endoderm
Epiblast → Mesoderm
Epiblast → Ectoderm

Endoderm

- Epithelial lining of Gut and associated glands (liver, pancreas, gall bladder)
- Epithelial lining of the respiratory organs (trachea, bronchus, lungs), tonsils, middle ear and pharyngotympanic tube.
- Parathyroid and thyroid glands
- Epithelial lining of bladder, most of urethra and lower vaginal canal

Mesoderm

Mesoderm → Lateral
Mesoderm → Intermediate
Mesoderm → Paraxial

Lateral

- Heart
- Spleen
- Adrenals
- Parietal / Somatic Layer
- Visceral / Splanchnic Layer

Paraxial

- Sclerotome - bone
- Myotome - muscle
- Dermatome - skin

Intermediate

- Urogenital organs and ducts

Ectoderm

Ectoderm →
- Epidermis, nails, sweat glands, lens, anterior pituitary
- Sensory cells of the ear and olfactory epithelium, lower anal canal

Ectoderm → Neuroectoderm

Neuroectoderm

Neuroectoderm → Neural Tube
Neuroectoderm → Neural Crest Cells

Neural Tube

- CNS
- Retina
- Pineal body
- Posterior pituitary

Neural Crest Cells

- PNS
- Melanocytes
- Glial & Schwann cells
- Cranial nerves
- Odontoblasts
- Connective tissue & bones of face & skull
- Cells of the adrenal medulla
- Bulbar and truncal ridges of the heart

The journey of development

- Fertilization to gastrulation - Some primordial cell differentiation happens - **Weeks 1-3**

<u>Organogenesis – **Weeks 4 to 8**</u>

Everything happens synchronously with many simultaneous events

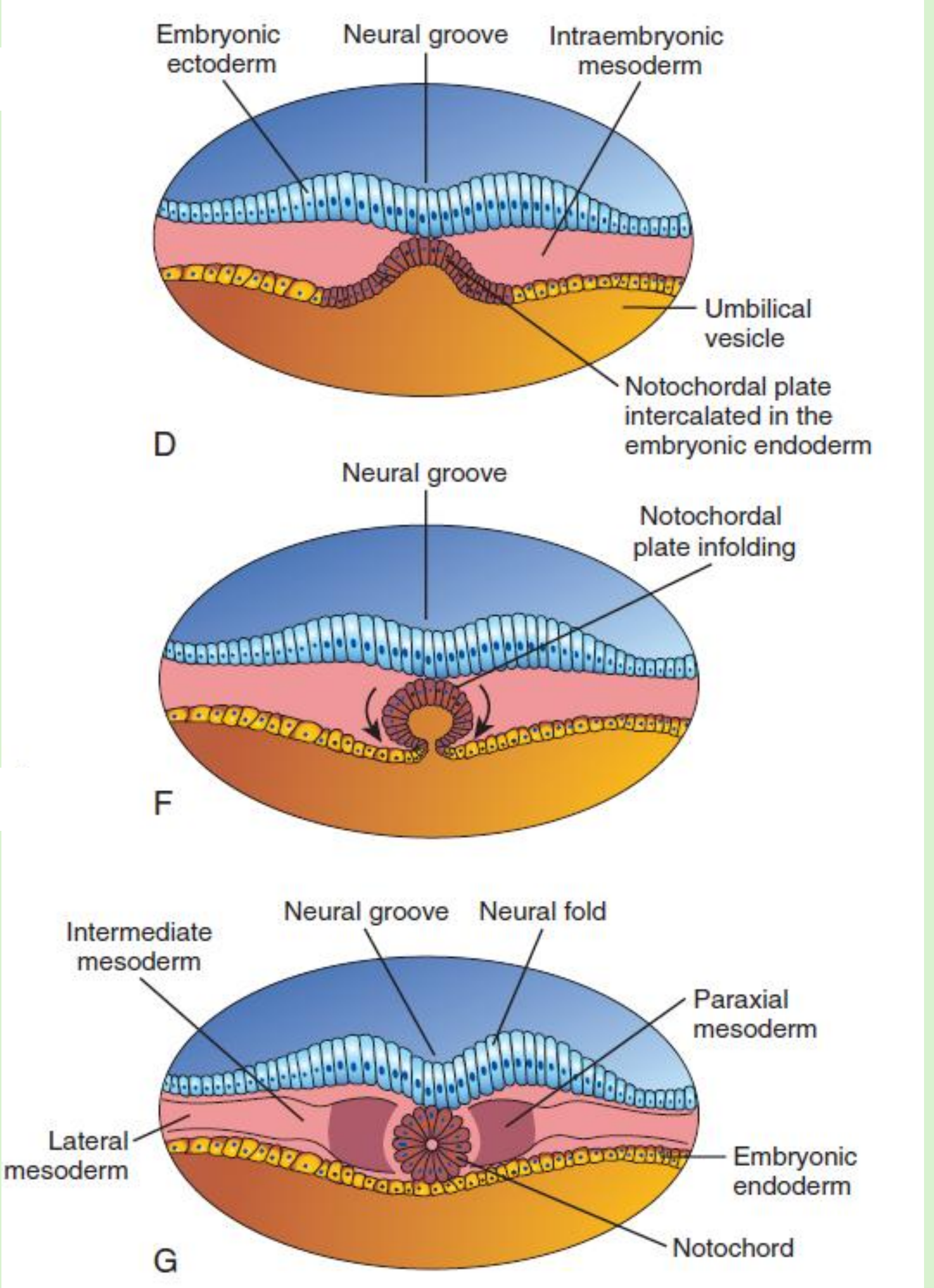
- Formation of the neural plate – day 18 – 21 **(Week 3)**
- Tissues start to differentiate into organ primordia – day 22 - 28
- Folding turns the embryo from a disc into a tube-inside-a-tube rendering its 3D form – day 22 – 28
  - Lateral folding turns the endoderm into the gut tube
- Paraxial mesoderm differentiate into somites - day 22 – 30
  - Somites start differentiating into the 3 components – day 24 - 30
- Formation of the neural tube – Neurulation – day 22 – 30
- Neural crest cell migration – Day 22 - 30
- Development of the spinal cord and nerves – Day 24 - 31
- Development of the vertebrae – Day 30 - 33

Week 4

Week 5

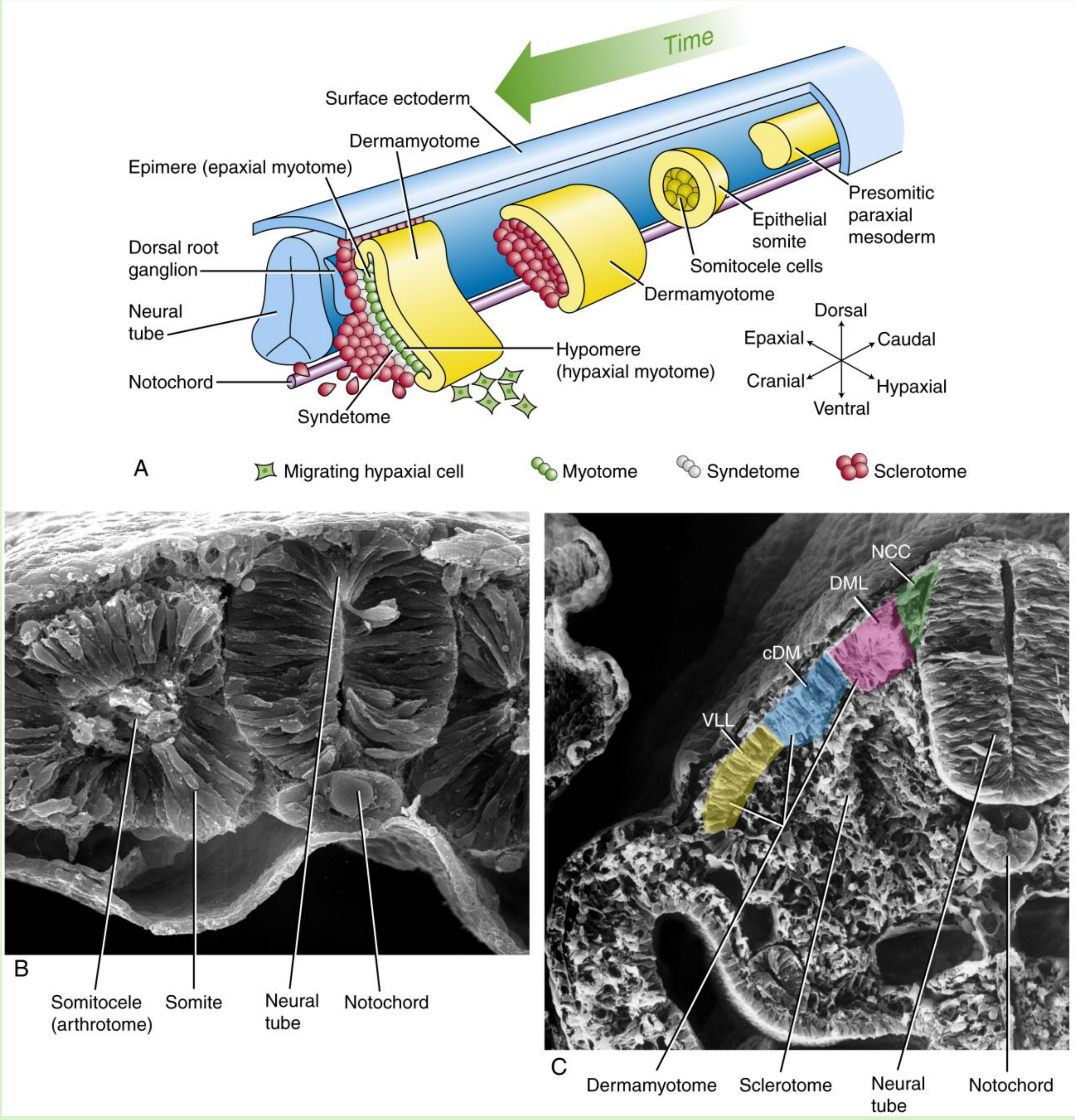
Recap: Week 3 – Notochord

- A solid cord of cells
- **Serves as the longitudinal axis of the embryo**
- Induces the overlying ectoderm cells to become elongated = **Neural plate (neuroectoderm)**
- **Induces differentiation of the mesoderm into distinct cell populations**
- **Contributes to the intervertebral discs (which remains in the adult)**



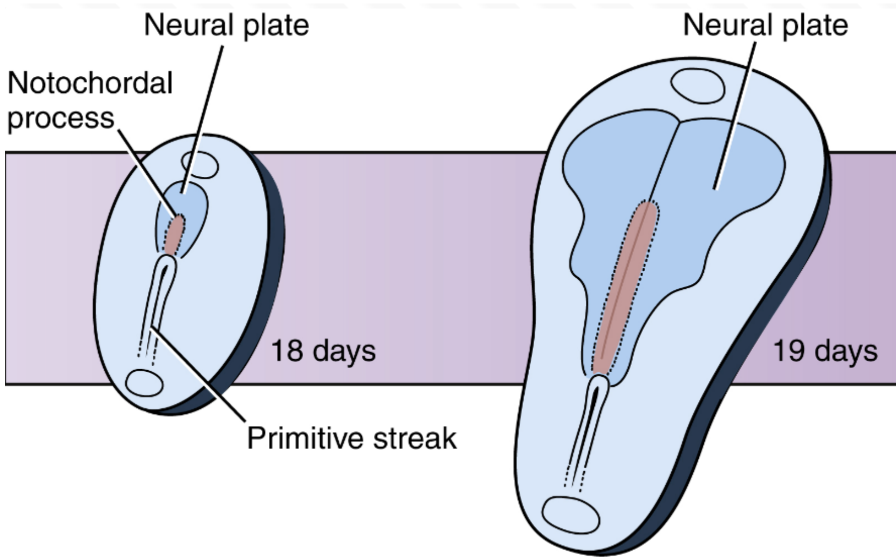
Recap: Week 3 – Paraxial mesoderm

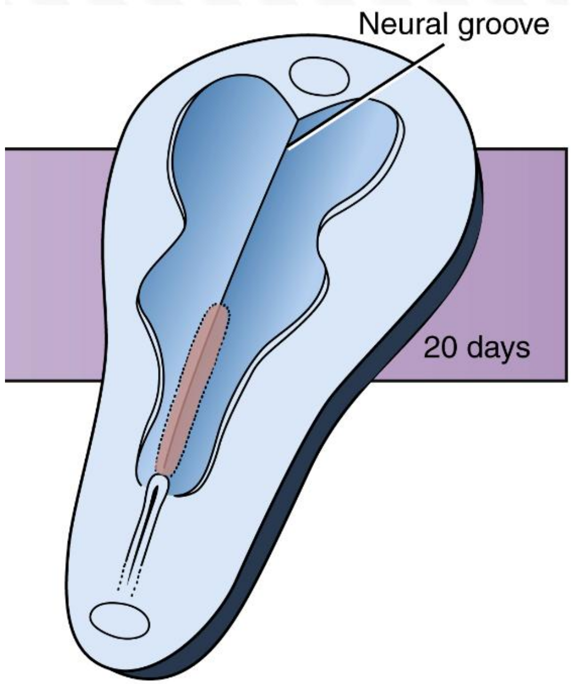
- The paraxial mesoderm form distinct bodies = **somites**
    - **Epithelial balls with central somitocele (loosely arranged cells)**
- Identified as bumps on the surface of the developing embryo
- Each somite has three distinct portions
    - **Sclerotome - bones**
    - **Myotome – skeletal muscle**
    - **Dermatome – dermis of skin**
- Each somite is associated with a spinal nerve



Week 3, day 18 – 21 Neurulation

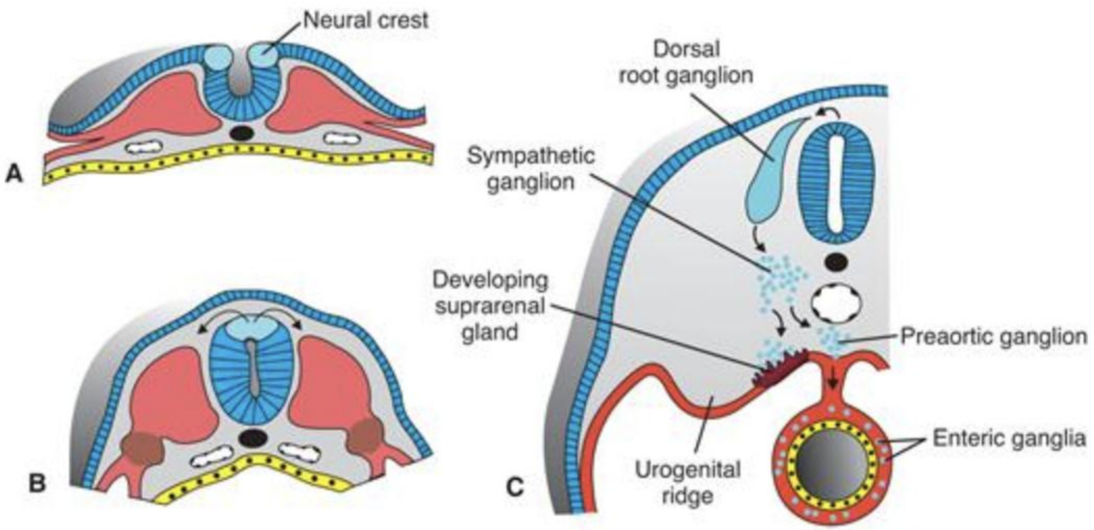
- Gastrulation continues on ventral surface
- Ectoderm differentiates into **neuroectoderm**
  - **Cranial** to the primitive streak, on the **dorsal surface** of the embryonic disc
  - Slipper shaped elevation - **Neural plate**
- Changes are induced by the primitive streak and notochord
  - Many intercellular signalling molecules are important
- Central indentation - **neural groove**
  - Influenced by developing mesoderm and the position of the notochord

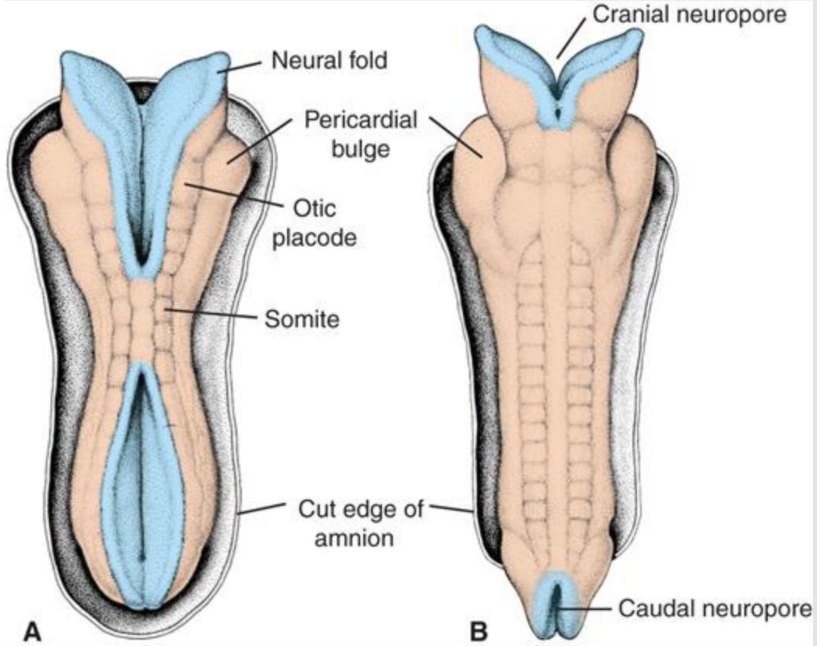




Week 4, day 22 – 28 Neural tube formation

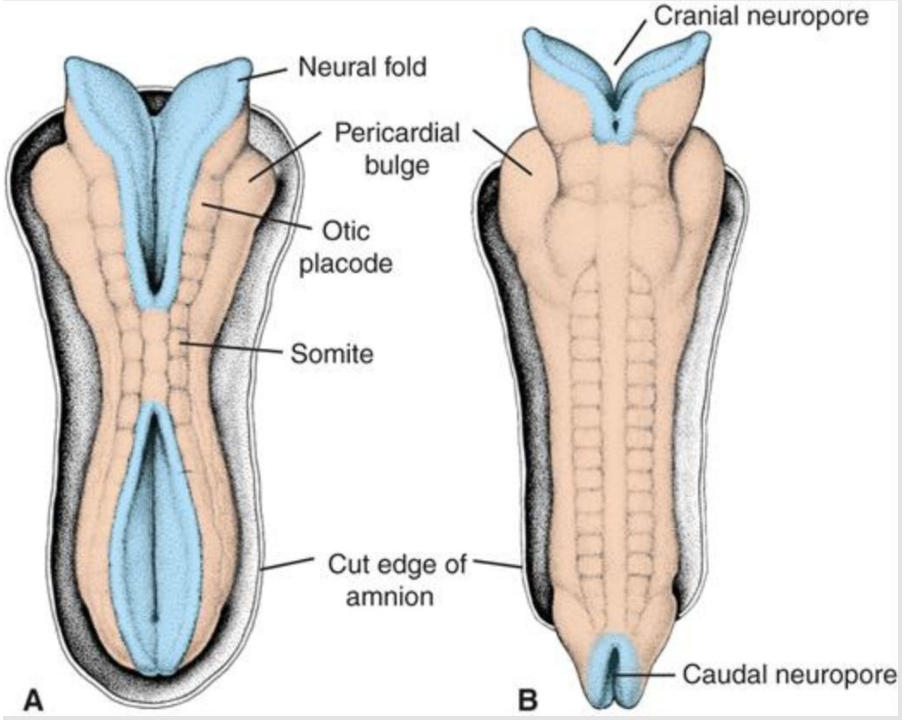
- The lateral edges of the neural plate fold toward the midline
  - Due to differential growth and condensation of the somites
- As the crests of the neural fold approach each other the cells at the tip break away and differentiate into mesenchyme – **Neural crest**
  - split off the neural tube as it closes
  - starts in the cervical region
  - migrates toward specific regions

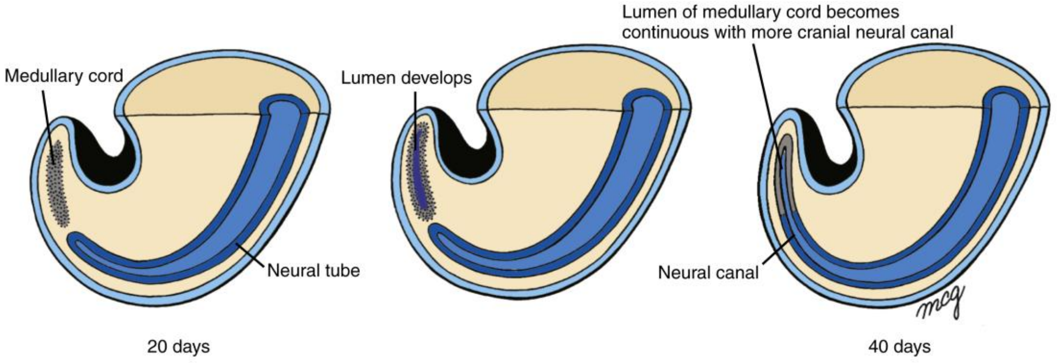




Week 4, day 22 – 28 Neural tube formation

- Closure of the neural tube is **segmental**
  - **Starts in the neck** region and moves cranially and caudally
  - Leaves the cranial and caudal (somite 31 location) neuropores
    - **Cranial neuropore closes day 25**
    - **Caudal neuropore closes day 27/8**
- The most caudal end **(sacral and coccygeal segments)** of the neural tube, neural crest cells, and somites develop from the **tail bud**
  - condensation of central tail bud cells into a solid mass called the **medullary cord**
  - undergoes cavitation to form a lumen, and merges with the neural canal





Neuroectoderm derivatives

Neuroectoderm

Neural Tube

- CNS
- Retina
- Pineal body
- Posterior pituitary

Neural Crest Cells

- PNS
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- Odontoblasts
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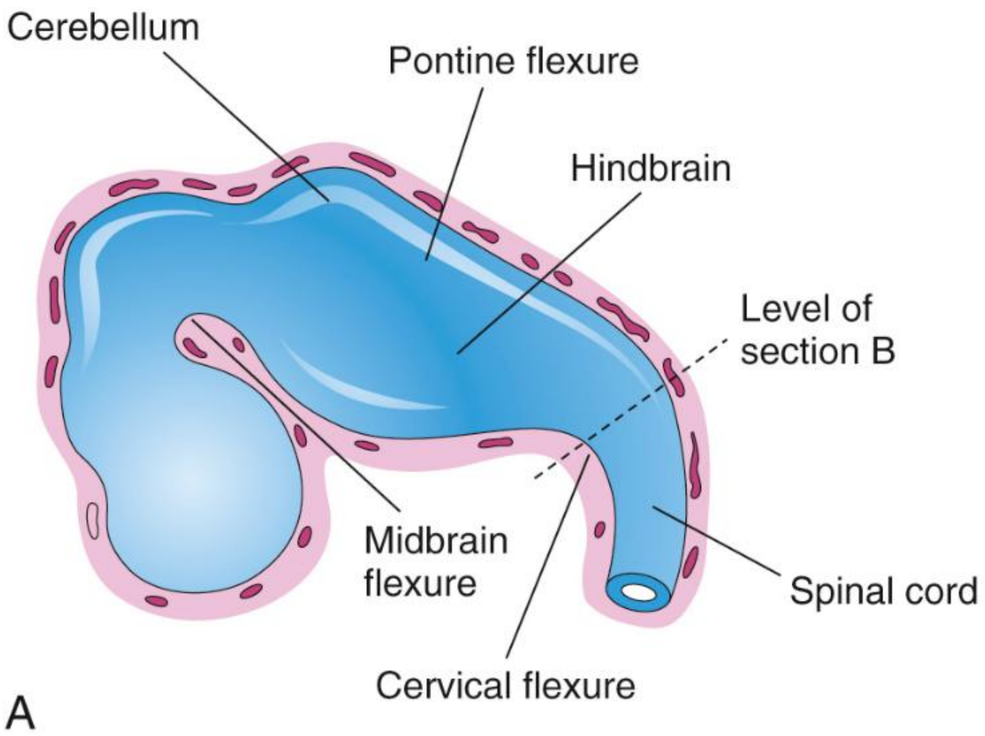
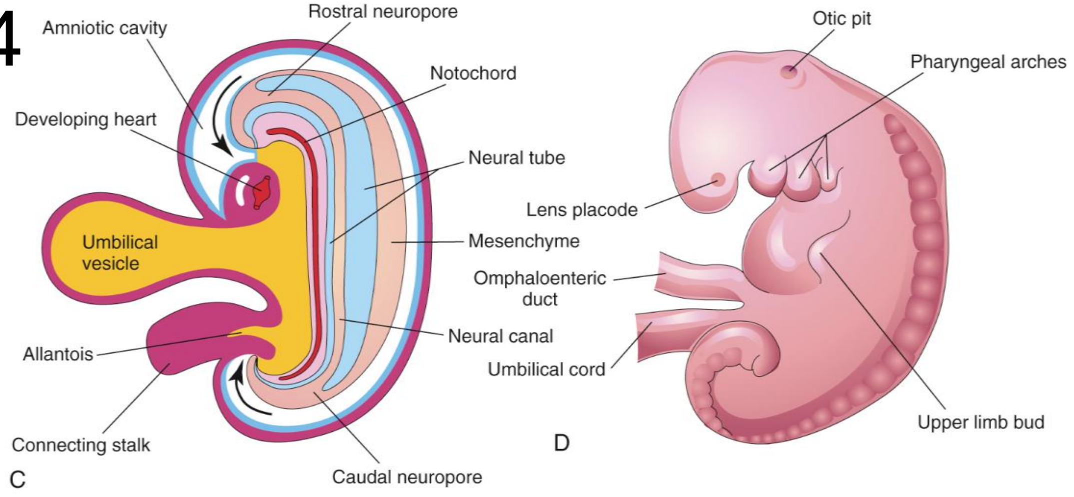


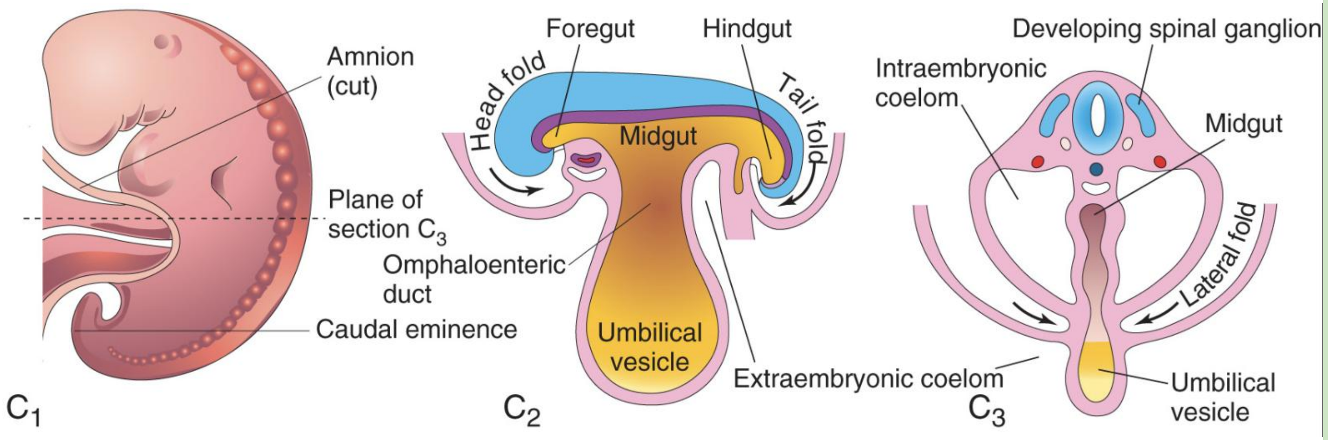
- The canal found inside the neural tube will become:
  - Ventricles
  - Central canal of spinal cord

Final form of the neural tube consists of neuroepithelium surrounded by mesenchyme

Meanwhile ....
Other things that happen in week 4

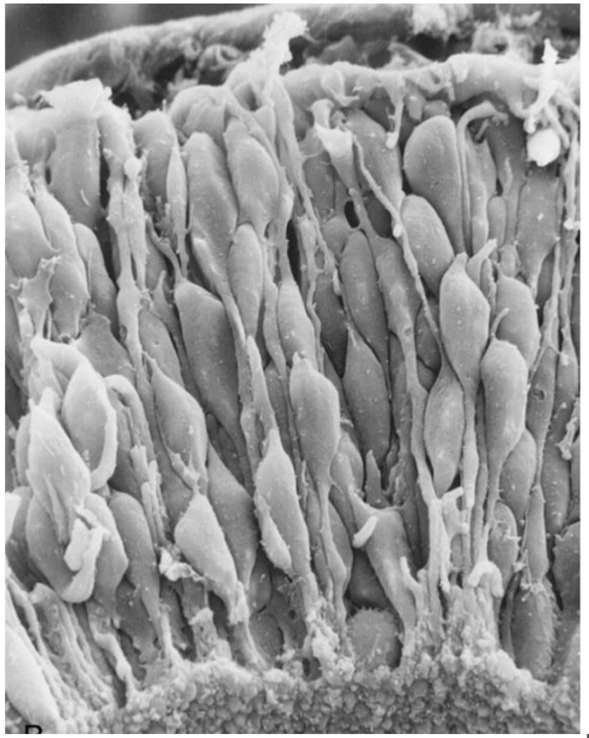
- Development of several organs initiate
  - Heart
  - Digestive tract
  - Eyes
  - Limbs
- More importantly the embryo undergoes folding

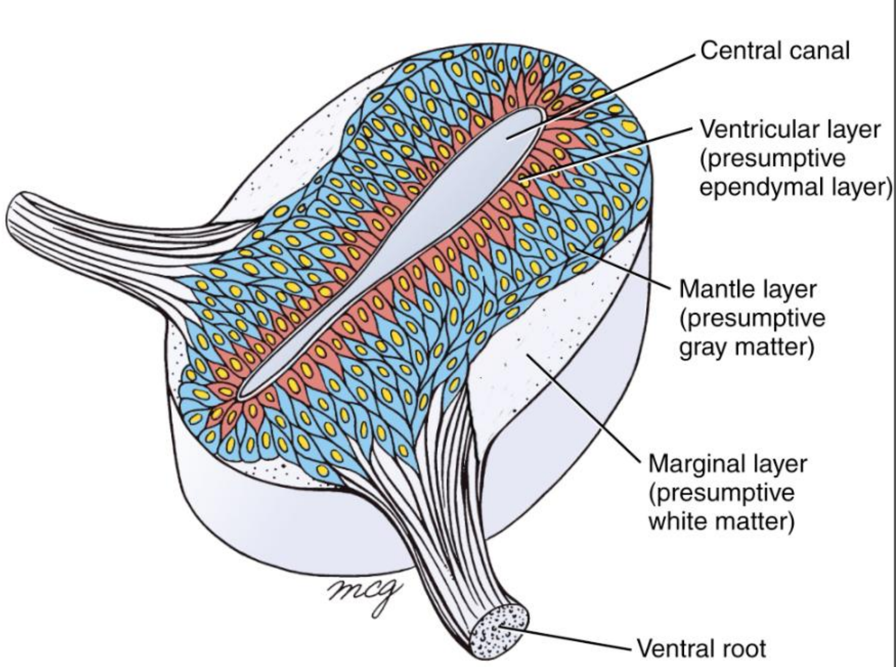




Neuroepithelial differentiation

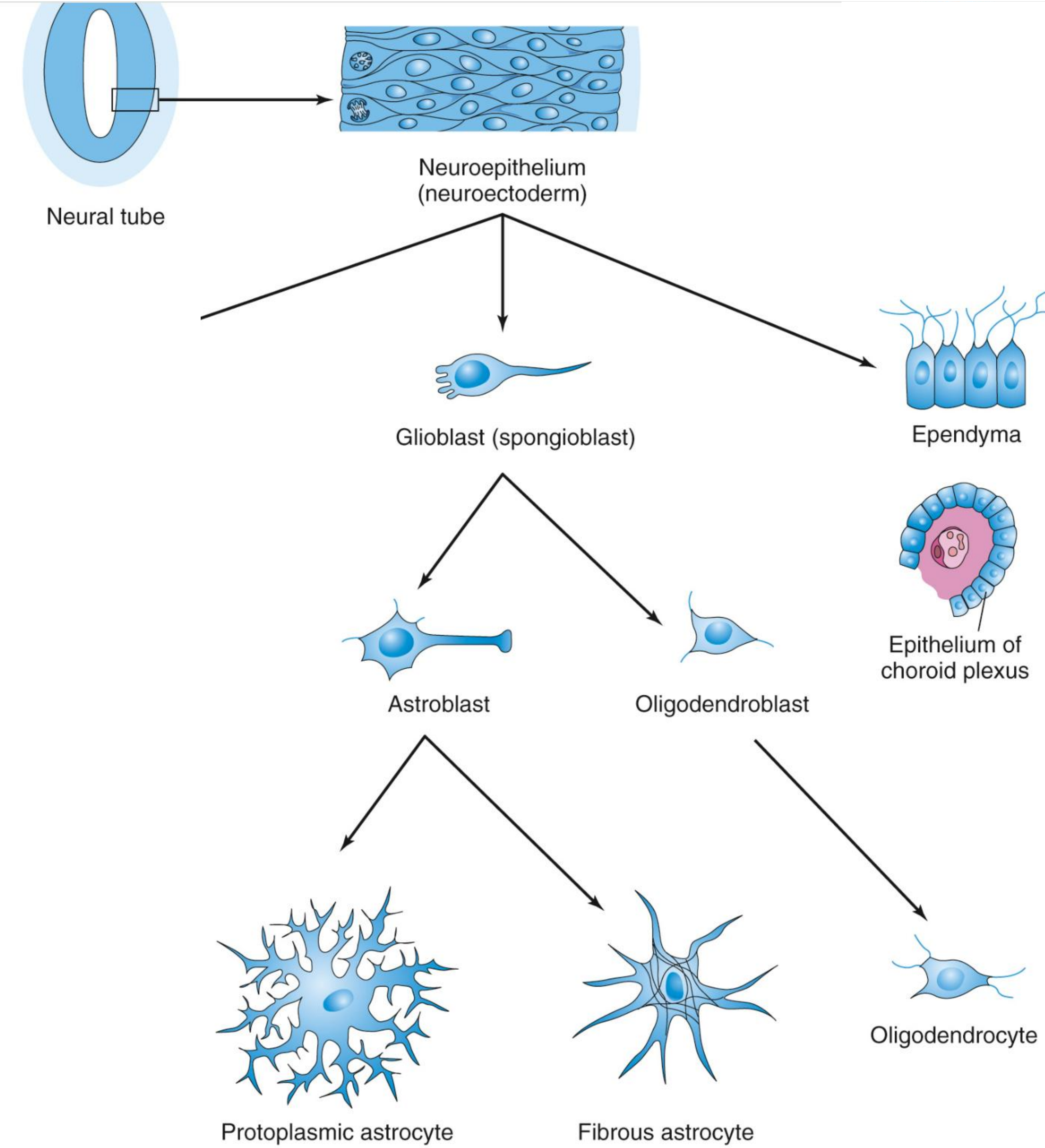
- From **neuroectoderm**
  - Pseudostratified columnar epithelium
  - Mitosis occurs at the luminal surface **(ventricular layer)**
- 1st – young neurons
  - Migrate to the periphery – **Mantle layer**
    - Forms the grey matter (cell bodies)
  - Processes extend further peripherally – **Marginal layer**
    - White matter (axons)
- Continual addition of new cells to the spinal cord results in eventual narrowing of the central canal
- Differentiation of the neuroepithelium into other cell types occurs along the same pathway as the closure of the neural tube





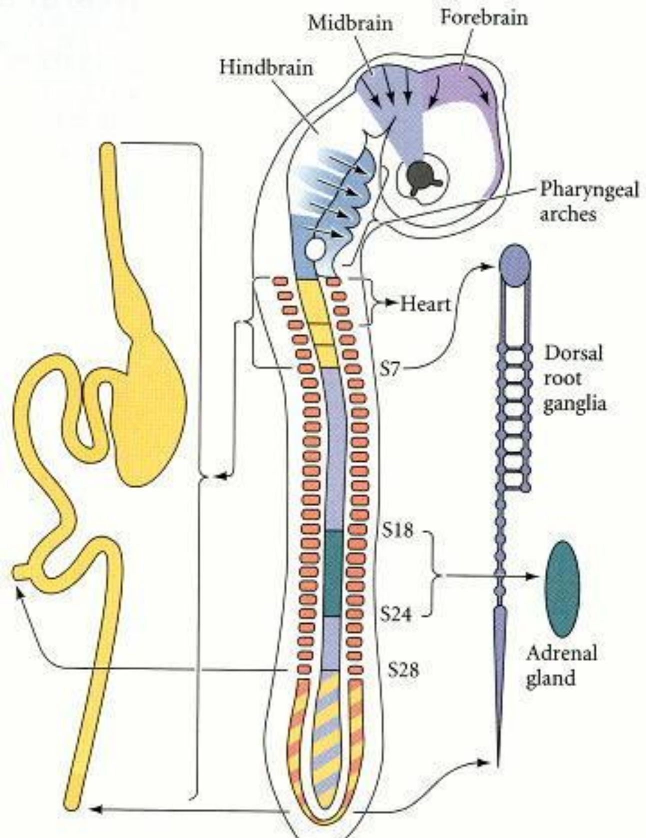
Neuroepithelial differentiation cont.

- Following a decline in neuron development
- **Glioblasts arise**
  - Astrocytes – blood brain barrier
  - Oligodendrocytes – myelin
- Remaining neuroepithelium differentiate into **ependymal cells**
  - Line the ventricular system and central canal
  - Contributes to formation of choroid plexus



Neural crest cells

- Arise along the entire length of the neural folds, bilaterally
- Each region gives rise to different tissues
  - **cranial (cephalic) neural crest** – migrate to pharyngeal arches
    - cartilage, bone, cranial neurons, glia, and connective tissues of the face
    - thymic cells, odontoblasts of the tooth primordia, and the bones of middle ear and jaw
  - **trunk neural crest –** have two migration pathways
    - dorsolaterally to form melanocytes of skin
    - ventrolaterally through cranial part of the sclerotome
      - pass through and continue ventrally = sympathetic ganglia, adrenal medulla and nerves around the heart
      - remain in sclerotome = dorsal root ganglia
  - **vagal (opposite somites 1-7) and sacral neural crest (somite 28)**
    - parasympathetic ganglia of the gut
  - **cardiac neural crest (Somites 1-3?)**
    - melanocytes, neurons, cartilage, and connective tissue or 3rd to 6th pharyngeal arches
    - entire musculoconnective tissue wall of the large arteries
    - the septum that separates the pulmonary circulation from the aorta



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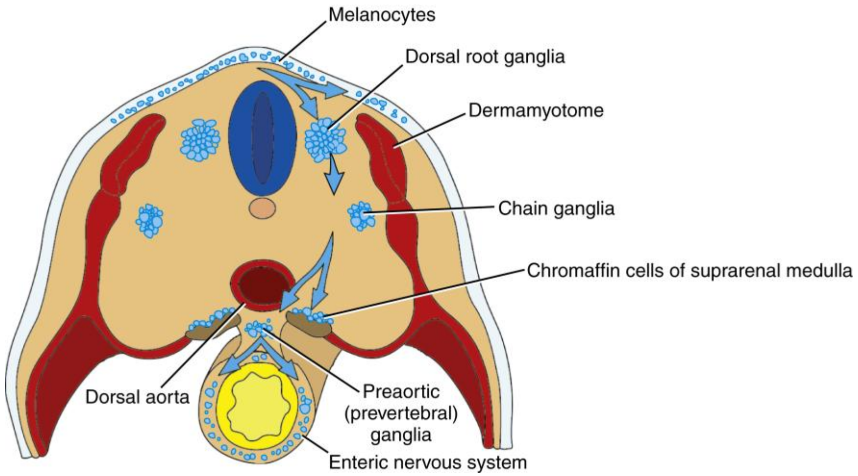
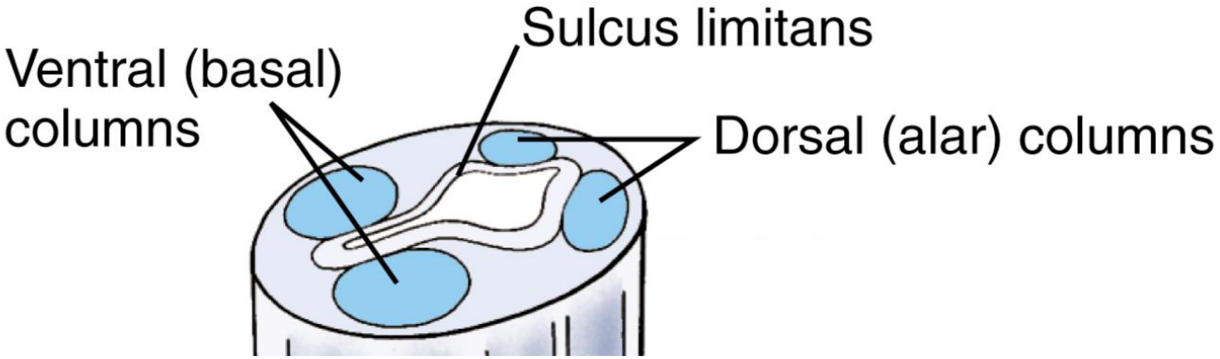
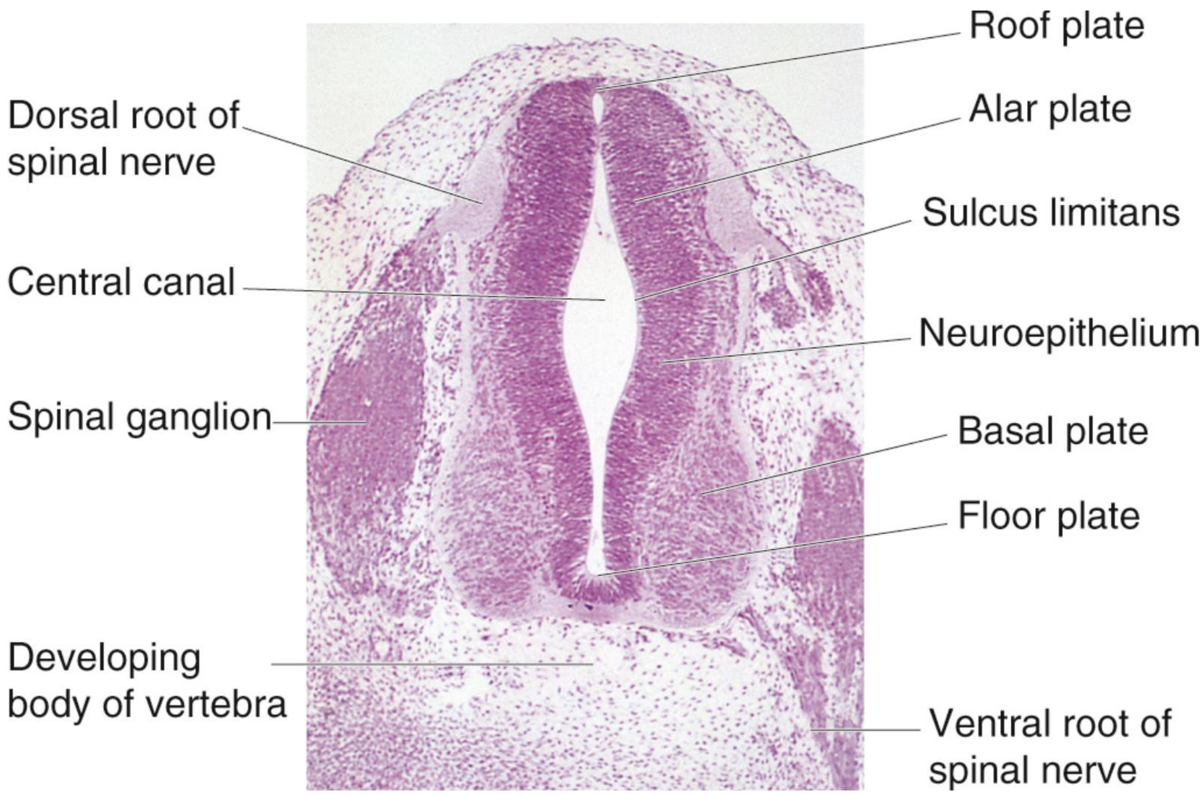


Week 4, day 27/28 – Development of the spinal cord

- Highly dependent on retinoic acid (Vit A)
- Develops from the caudal part of the neural tube
  - From the 4th somite level
- Differentiation of neuroectoderm causes four longitudinal bulges (collections of neurons) to form
  - Alar plate (dorsal)
  - Basal plate (ventral)
  - connected by thin roof and floor plates (non-neural tissue)
- Differential thickening causes the formation of grooves
  - Sulcus limitans
  - Separates the basal from the alar



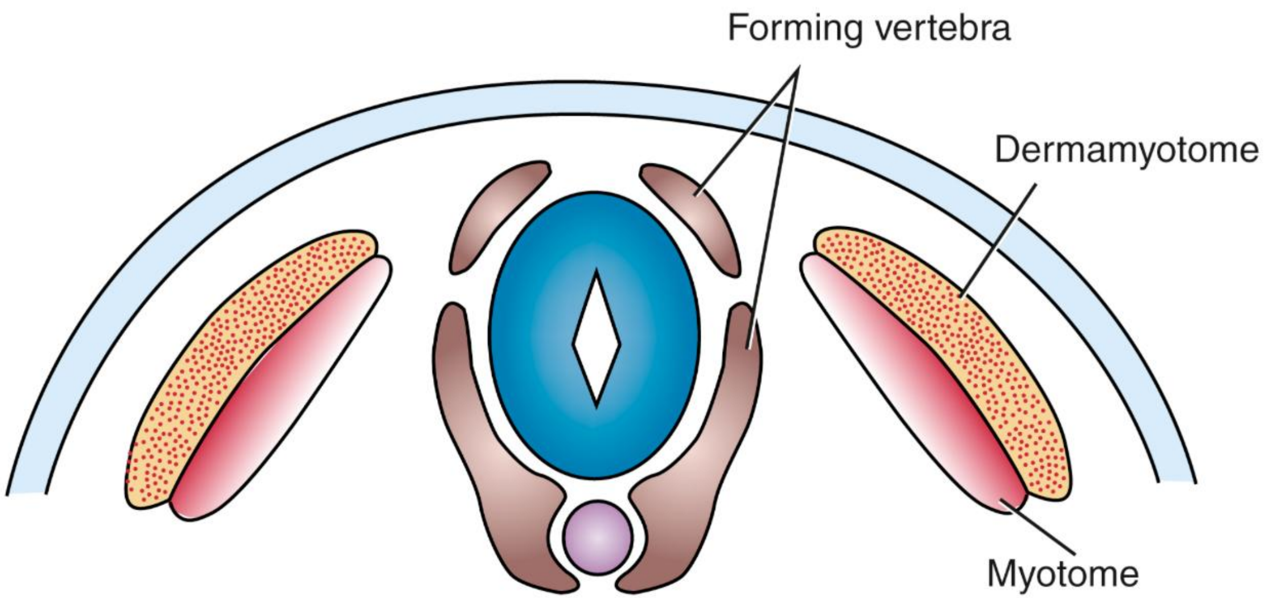
Dorsal



Ventral

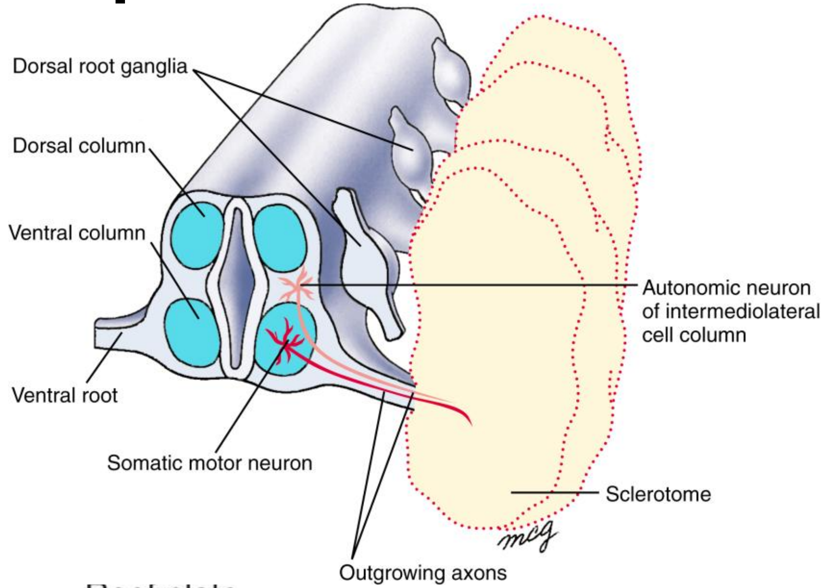
Week 5, day 30 - Development of the vertebrae starts

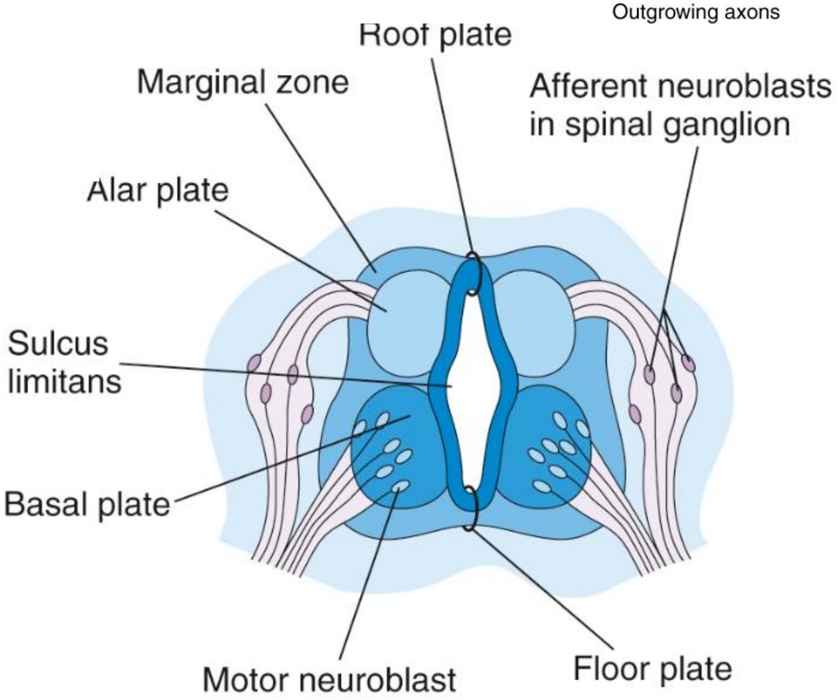
- Sclerotome of the somite becomes mesenchyme
  - Subdivided into cranial and caudal portions
    - By intrasegmental boundary – transverse layer of cells
  - Results in segmentation of neural crest and spinal nerves
- Ventral portion migrates to surround the notochord **– vertebral body**
- Dorsal portion migrate to surround the neural tube **– vertebral arch and spinous process**
- Lateral portions will become **transverse processes and ribs**



Week 5, day 30 - development of the spinal nerves

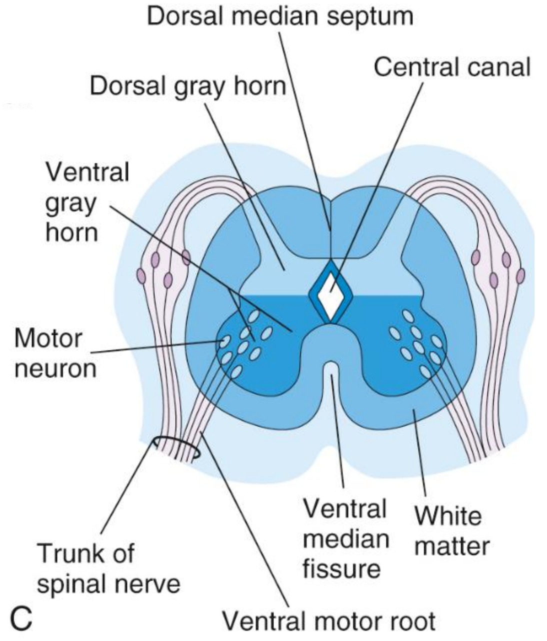
- **Axons** from the nerves developing in the **ventral horn** grow toward the **cranial sclerotome**
  - Starts in the neck
  - Initially a continuous broad band
  - Condenses into segmental nerves as they reach the cranial sclerotome
- Neural crest migrates to cranial sclerotome = **dorsal root ganglion**
- **Neurons in the DRG sprout axons in two directions – dorsal roots**
  - Central process synapses with developing neuron in dorsal horn
  - Peripheral process grows toward organs

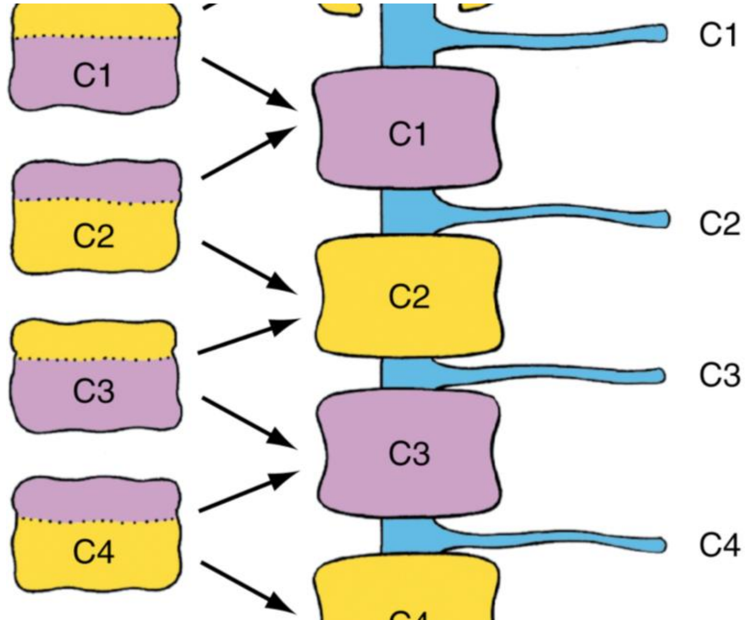




Week 5 – 9 – Development of spinal cord, nerves and vertebrae cont.

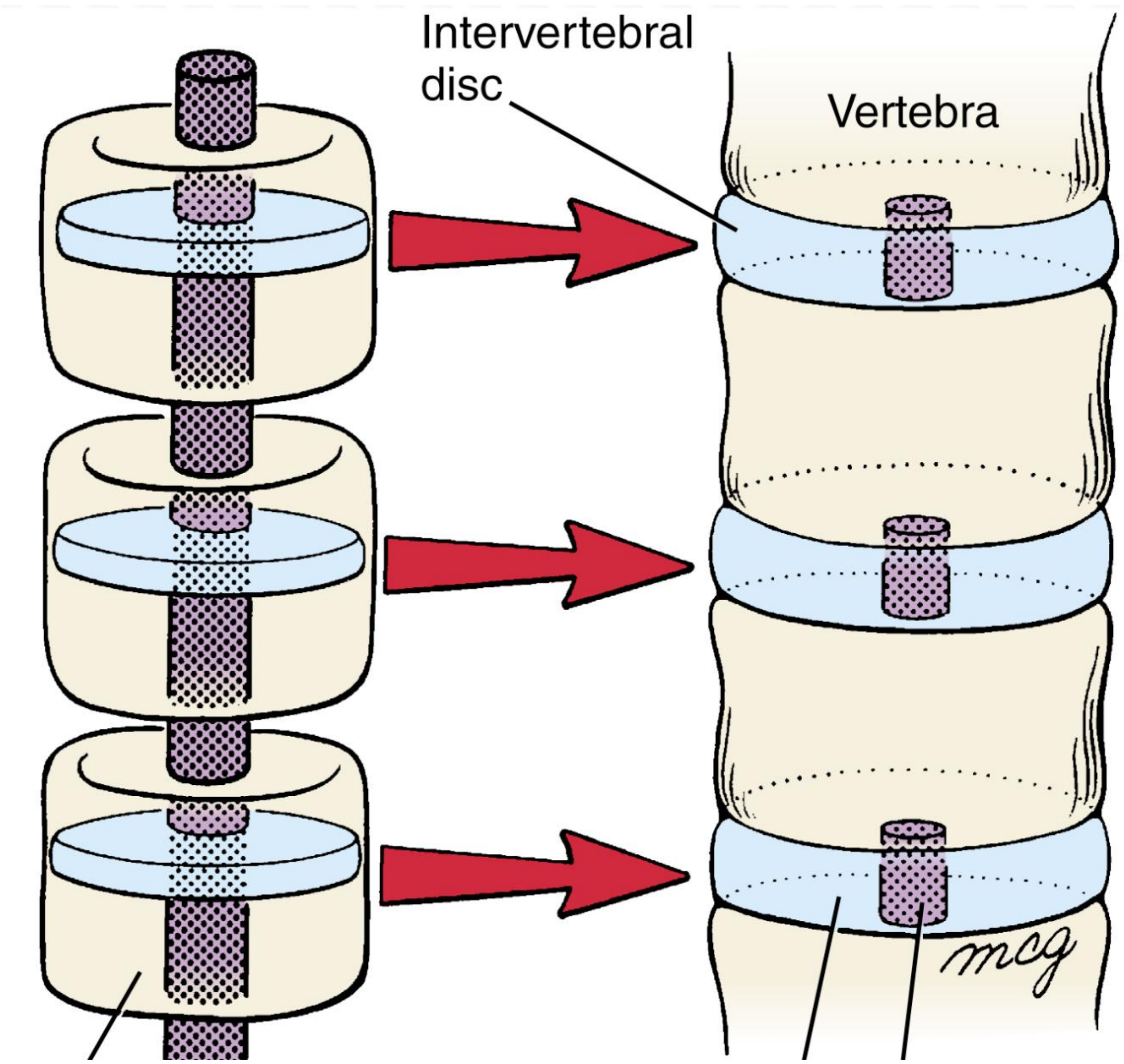
- Sclerotome condenses around the notochord
- Splits along the intrasegmental boundary = **resegmentation**
- Cranial portion of first cervical somite contributes to occipital bone
- The remaining segments form by fusion of the caudal portion of one vertebra to the cranial portion of the one below
- Causes the nerves to pass between the vertebral bodies
- Spinal cord continues to grow by addition of new cells
- Central canal becomes tiny





Week 5 – 9 – Development of the intervertebral disc

- Develops at the intrasegmental boundary
- Notochord enlarges in these segments to form the gelatinous nucleus pulposus
  - degenerates at the sites of vertebral body development
- Anulus fibrosis forms from the somitocele



Summary

- Development of neural tube depends on notochord and paraxial mesoderm
- Neural crest cells migrate to various locations
- Somites differentiate into
  - Sclerotome
  - Myotome
  - Dermatome
- Nerve axons and neural crest migrate toward the cranial sclerotome to form spinal nerves
- Vertebrae develop from sclerotome around the notochord and neural tube
- Intervertebral discs form at the sites of resegmentation
- Notochord becomes the nucleus pulposes

Watch these videos for visual representation

- https://premiumbasicsciences.lwwhealthlibrary.com/multimediaplayer.aspx?MultimediaID=19450545